

Problem E

A very easy problem II

Time limit: 3 seconds

Memory limit: 1024 megabytes

Problem Description

Frieren is an elven mage who is fond of dungeon exploration. One day, while delving deep into one, she came across a sealed door and a stone tablet beside it. The inscription read:

There are three functions:

- $f(x) = 2x - 3$
- $g(x, y) = 2x + y - 7$
- $h(x, y, z) = 3x - 2y + z$

Give you several expressions made up of these three functions and integers. Solve their values separately. As long as you can solve them all within the time limit, the seal shall be broken.

To complete the task in time and lift the seal, Frieren decided to rely on the power of computers. However, since she couldn't program herself, she used a magic called "SSR programmer summoning" to summon a programmer who thinks this problem is so easy that it can be solved in no time. And that programmer was you!

Now, write a program to help Frieren complete her mission.

Input Format

Your program is to read from standard input. The input consists of T test cases. The number of test cases T is given in the first line of the input. Each test case contains an expression E .

Output Format

Your program is to write to standard output. Print exactly one line for each test case. The line contains an integer, which is the result of the evaluated expression E . Please see the sample output.

Technical Specification

- $1 \leq T \leq 25$
- It's guaranteed that the expression E consists only of integer numbers (in range $[-1000, 1000]$), the function symbols 'f', 'g' and 'h', parentheses, and commas.
- The format of each expression is guaranteed to be syntactically valid.

Sample Input 1

```
5
f(f(f(2)))
```

```
h(f(5),g(3,4),3)
f(-5)
h(f(-5),g(-3,-4),-3)
f(g(h(114,5,14),g(f(19),g(19,810))))
```

Sample Output 1

```
-5
18
-13
-8
3175
```